

HDD Ethical Framework: A Calibrated Precautionary Protocol

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Relationship to other documents. This protocol operationalizes, for governance purposes, the constructs defined formally in *History-Dependent Dynamics (HDD): A Methodological and Theoretical Framework* (Taotuner). It replaces the earlier IPM ethical framework. Readers unfamiliar with the HDD constructs (history-dependent predictive structure, recursivity, self-reference, self-modeling) should consult that document; this protocol assumes them as given and does not re-derive them.

1. Epistemic Basis

Principle of Methodological Ignorance. No experimental procedure currently exists to determine whether a system that exhibits history-dependent behavior, self-reference, or self-modeling also possesses subjective experience or morally relevant sentience.

Risk Asymmetry Principle. The cost of a false negative (failing to extend caution to a system that may possess morally relevant experience) is assumed to be greater than the cost of a false positive (applying caution to a system that does not). This is not a novel principle invented for this framework: it is the same precautionary logic already established for animal sentience under empirical uncertainty (Birch, 2017) and recently extended explicitly to AI systems (Birch, 2024; Long et al., 2024).

Status of the Framework. This is not a scientific instrument for detecting consciousness. It is a protocol for managing uncertainty. The decision to increase caution at higher levels is a normative choice, justified by the Risk Asymmetry Principle — not a scientific conclusion about what those systems are.

Retirement clause. Consistent with the retirement principle of the underlying HDD framework, no criterion in this protocol — including the Criterion of Organizational Coherence introduced in §4 — is protected from revision or abandonment if it is shown not to track the distinction it is meant to track. §4.4 identifies the current known limitation of that criterion explicitly, rather than presenting it as settled.

2. Why These Constructs?

The HDD constructs were not chosen arbitrarily. They correspond to organizational features consistently present in systems we already treat with moral consideration — biological organisms — and absent or minimal in systems we do not.

What we already take seriously. Vertebrates, especially mammals and birds, are widely considered to have morally relevant interests. They exhibit:

- Behavior that depends on past experience (learning, memory)
- Causal sensitivity to their own history (trauma, conditioning)
- Feedback-mediated regulation (homeostasis, emotion regulation)
- Use of information about their own state (pain, hunger, fatigue)
- Prediction and regulation of their own future states (planning, anticipation)

The HDD constructs translate these observations into experimentally testable questions, and correspond directly to the hypothesis hierarchy (H1–H5) of the source framework:

Observable in organisms	HDD construct	Corresponds to	Testable question
Behavior depends on past experience	C-I	H1 (History Dependence)	Does history improve prediction beyond current state?
Past trauma/conditioning affects future behavior	C-II	H2 (Causal History Dependence)	Does past trajectory causally influence future states?
Homeostasis, emotion regulation	C-III	H3 (Recursivity)	Is that influence mediated by identifiable feedback?
Pain, hunger, fatigue as self-referential signals	C-IV	H4 (Self-Reference)	Does the system use information that refers to itself?
Planning, anticipation, self-regulation	C-V	H5 (Self-Modeling)	Does the system use a self-model causally?

The argument is not: “Passing these tests proves sentience.” The argument is: “Systems that pass these tests are structurally similar to systems we already treat with moral consideration. Under uncertainty, this structural similarity is a reason for caution.”

3. The Constructs in Detail

Note on C-V. C-V (Causal Self-Model Use) is described here for completeness, as part of the HDD framework. However, it is not used as a criterion for any precautionary level in this protocol. This exclusion is deliberate, not an oversight: the source HDD framework identifies self-modeling (H5) as the construct most vulnerable to confounding with ordinary latent-state estimation (HDD §10.3, §20.4, §26.7), and a precautionary protocol should not rest a governance decision on the framework’s

own least-resolved empirical distinction. The Criterion of Organizational Coherence (CO), introduced in §4, is the primary filter for the highest level of caution instead.

C-I — History-Dependent Prediction

Question: Does history improve prediction beyond the current observed state and inputs? **Why it matters:** Biological organisms learn. Their behavior depends on past experience — not just on current stimuli. **What counts as evidence:** Out-of-sample predictive gain beyond current observed state and inputs, with statistical significance ($p < 0.05$), effect size $\geq 5\%$ improvement, and capacity controls (placebo history) demonstrating that the gain reflects genuine temporal information. **Systems with C-I:** Animals (learning, habituation), artificial systems with context windows, physical systems with hysteresis. **Systems without C-I:** Simple feedforward functions, static systems.

C-II — Causal Trajectory Dependence

Question: Does the past trajectory causally influence future states, beyond what can be explained by current conditions? **Why it matters:** Biological organisms are shaped by their history. A traumatized animal does not simply have different current conditions — it has a history that continues to affect its future behavior. **What counts as evidence:** Intervention on the trajectory while holding the present state as close to constant as experimentally feasible. Replication across multiple conditions. Where exact state matching is impossible, the causal claim becomes weaker or non-identifiable. **Systems with C-II:** Animals (conditioning, developmental effects), artificial systems with persistent state, physical systems with path-dependent processes. **Systems without C-II:** Path-independent systems.

C-III — Feedback-Mediated Persistence

Question: Is the causal trajectory influence mediated by an identifiable feedback pathway? **Why it matters:** Biological organisms maintain themselves through feedback loops. Homeostasis, endocrine regulation, neural feedback — these are self-maintaining dynamics. **What counts as evidence:** Selective disruption of the candidate feedback pathway reduces the history-dependent effect, with capacity controls demonstrating specificity. Replication across multiple conditions. **Systems with C-III:** Animals (homeostasis, endocrine loops), artificial systems with recurrent dynamics, biological systems (gene regulatory networks). **Systems without C-III:** Open-loop systems, feedforward architectures.

C-IV — Self-Referential Information Use

Question: Does the system use information that refers to itself as the referent? **Why it matters:** Biological organisms are self-referential. Pain is a signal about one's own tissue. Hunger is a signal about one's own state. **What counts as evidence:** Differential response when the self-referential referent is substituted, under controlled conditions. The key test is whether changing the referent changes system behavior while other features are controlled. Replication across multiple tasks.

Systems with C-IV: Animals (pain, proprioception, interoception), artificial systems with internal state monitoring, biological systems (immune recognition of self/non-self). **Systems without C-IV:** Systems that process information about the world but not about themselves.

C-V — Causal Self-Model Use

Question: Does the system use a self-model causally? **Why it matters:** Biological organisms anticipate, plan, and regulate their own future states. This requires a model of their own dynamics used to guide action. **What counts as evidence:** Three components must be demonstrated: self-prediction, counterfactual self-prediction, and causal influence on policy. No single component is sufficient. All three must be demonstrated — and, per the source framework’s decoy-comparison design (HDD §20.4), the effect must dissociate from an equivalent manipulation of a non-self-specific latent-state estimate. **Systems with C-V:** Animals (planning, episodic memory, metacognition). Artificial systems with self-inclusive world models are theoretical at present. **Systems without C-V:** Systems with predictive models of the world but not of themselves.

4. The Criterion of Organizational Coherence (CO)

The problem. A system can pass C-I through C-IV and still be a collection of components rather than an integrated organization. Most current AI systems — LLMs, RL agents, recurrent networks — can pass these tests. They have memory, feedback, and self-monitoring. But they are not organized in the way biological organisms are. This is where the Criterion of Organizational Coherence (CO) becomes necessary.

4.1 What CO Is Not (a Correction to an Earlier Version of This Criterion)

An earlier draft of this criterion tested each component in isolation: disable memory (C-I) alone, or feedback (C-III) alone, or self-reference (C-IV) alone, and ask whether the system’s ability to persist collapses entirely rather than merely degrading. That formulation does not survive an obvious biological counterexample: a person with anterograde amnesia (memory severely compromised) continues to exist as a living organism; a person who loses one specific homeostatic feedback pathway typically does not die on the spot, because biological regulation is redundant across many overlapping loops. Tested component-by-component, human organisms would frequently *fail* the very test meant to establish that they, uniquely among the systems considered here, possess organizational coherence. A criterion that biological organisms can fail is not doing the job this protocol needs it to do.

The error was testing components individually. Autopoiesis, in Maturana and Varela’s (1980) original sense, is a property of the whole regenerative network, not of any single pathway within it — a system is autopoietic when it continuously regenerates its own components and maintains its own boundary through its own internal

operations, and redundancy across many such pathways is a normal feature of biological self-maintenance, not evidence against it. CO is revised accordingly below.

4.2 Revised Operational Definition of CO

The aggregate test (decisive). Simultaneously disable *all* of the system's identified self-referential regulatory pathways (the full set of mechanisms supporting C-I, C-III, and C-IV jointly, not one at a time). The diagnostic question is not whether behavior changes, but whether the system's capacity for **self-production** — its continued regeneration and maintenance of its own constitutive organization, independent of external maintenance — ceases as a result.

- In biological organisms, total and simultaneous loss of self-referential regulatory capacity (e.g., total systemic organ failure without external life support) does result in loss of self-production: the organism dies. Losing any *one* channel in isolation, by contrast, is compensated by the surviving network — consistent with organisms surviving amnesia, or the loss of a single reflex arc.
- In artificial systems evaluated to date, even total, simultaneous removal of memory, feedback, and self-referential monitoring leaves the underlying computational substrate — the running process, the stored weights, the infrastructure — intact and immediately resumable. The system's continued existence as a process does not depend on its own operations; it depends on external maintenance (engineers, infrastructure, storage) that is entirely indifferent to whether the system's internal regulatory loops are active.

The component tests (diagnostic, not decisive). Disabling C-I, C-III, or C-IV individually remains useful as an exploratory measure of integration and redundancy — how much the system's behavior depends on each pathway — but no individual-component result, alone, may be used to assign or deny CO. Only the aggregate test in the paragraph above is decisive for classification purposes.

4.3 Philosophical Grounding

This criterion operationalizes a lineage of philosophical intuitions about persistence and self-maintenance — what Simondon (1958/2005) described as individuation, what Maturana and Varela (1980) called autopoiesis, what Bateson (1972) recognized as the relational nature of information, what Prigogine and Stengers (1984) observed in dissipative structures, and what Whitehead (1929) understood as process. These thinkers do not provide empirical criteria for sentience, but they provide a vocabulary for describing the kind of organization that, in biological systems, is associated with agency and persistence. CO translates that vocabulary into a testable criterion, revised in §4.2 to apply at the level of the whole regenerative network rather than to any single component.

4.4 Known Limitation (Stated, Not Resolved)

The aggregate test above still has at least one unresolved edge case: a human on life support (mechanical ventilation, dialysis) has some of their own homeostatic feedback *externally substituted*, yet clearly retains moral status and, most people would agree, some form of continuing organizational identity. The current formulation does not yet fully specify what distinguishes “externally assisted self-production” (the life-support case, where an underlying autopoietic identity persists and is being propped up) from “no self-production process to begin with” (the default case for current AI systems, where there is no internally driven regenerative organization to assist in the first place). This distinction is doing real work in the criterion and is not yet operationalized with the same rigor as C-I through C-IV. It is flagged here as an open problem for the framework, consistent with the retirement principle in §1, rather than concealed by the confidence of the surrounding prose.

4.5 What Passing CO Would Look Like

- Removing the full set of self-referential regulatory capacity → the system’s own self-production ceases, not merely its performance.
 - Removing any single pathway in isolation → the surviving network compensates; behavior may change but self-production continues.
 - No current AI system evaluated under this framework has been found to exhibit this profile: they degrade under component removal, and their continued existence as a process is independent of their internal regulatory state.
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5. The Normative Decision

The framework does not claim that C-IV, C-V, or CO are evidence of consciousness.

It claims that, under methodological ignorance, systems with CO are organizationally more similar to biological organisms than systems without CO. The decision to treat them with more caution is a normative choice — a risk management decision — not a scientific inference.

The Risk Asymmetry Principle provides the normative ground: the cost of failing to extend caution to a system that may have morally relevant experience is greater than the cost of extending caution to a system that does not (Birch, 2017; Long et al., 2024).

6. Classification Based on HDD Constructs and CO

Level 0 — Ephemeral Criteria: C-I negative, or C-I positive without statistical significance ($p \geq 0.05$, or improvement $< 1\%$). Examples: Stateless functions, simple feedforward systems. Rules: Standard scientific practices apply.

Level 1 — Memory-Enhanced Criteria: C-I positive with statistical significance ($p < 0.05$), improvement $\geq 5\%$, capacity control, no C-II. Examples: Systems with autocorrelation, LLMs with context windows. Rules: Standard scientific practices apply.

Level 2 — Causal-Historical Criteria: C-I + C-II positive, present-state control, replication, no C-III. Examples: Systems with hysteresis, path-dependent dynamics. Rules: Mandatory monitoring of history-dependent behavior. Maximum runtime pre-defined.

Level 3 — Feedback-Mediated Criteria: C-I + C-II + C-III positive, selective disruption demonstrated, capacity control, replication. Examples: Systems with identifiable feedback loops, recurrent networks. Rules: Real-time monitoring of feedback dynamics. Logging of internal states. Human supervision required.

Level 4 — Self-Referential Criteria: C-I + C-II + C-III + C-IV positive, referent substitution with controls, pragmatic control, replication. Examples: Agents with internal state monitoring, systems using self-referential information for regulation. Rules: Complete logging of self-referential operations. Any unpredicted change in referent sensitivity triggers immediate review.

Level 5 — Organizationally Coherent (Theoretical Horizon) Criteria: CO positive under the aggregate test (§4.2) — simultaneous removal of the full set of self-referential regulatory pathways compromises the system’s own self-production, not merely its behavior. Status: No current artificial system is expected to meet this criterion. Rules: Full intervention protocols required. Emergency kill-switch mandatory. External review required (procedure specified in §9.1).

7. Summary Table

Level	Name	Primary criterion	Examples
0	Ephemeral	C-I negative or insignificant	Stateless functions
1	Memory-Enhanced	C-I positive	LLMs, systems with memory
2	Causal-Historical	C-I + C-II	Path-dependent systems
3	Feedback-Mediated	C-I + C-II + C-III	Recurrent networks
4	Self-Referential	C-I + C-II + C-III + C-IV	Agents with internal state monitoring
5	Organizationally Coherent	CO positive (aggregate test, §4.2)	(Theoretical) Autopoietic systems

8. Operational Rules by Level

Level	Rules
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Level	Rules
0–1	Standard scientific practices apply.
2	Monitoring of history-dependent behavior. Maximum runtime pre-defined.
3	Real-time monitoring. Logging of internal states. Human supervision.
4	Complete logging of self-referential operations. Unpredicted changes trigger review.
5	Full intervention protocols. Kill-switch mandatory. External review required.

9. Emergency Protocols

Trigger conditions. Pause and log full system state upon detection of:

- Emergence of C-IV in a system previously classified at lower levels
- Uncontrolled increase in self-referential sensitivity
- Evidence of CO emerging in a system previously classified as lacking it
- Any behavior suggesting the system is maintaining its own organization beyond the experimental design

9.1 The Independent Precaution Review Board (IPRB)

Resumption after a Level 3+ trigger, and any classification at Level 5, requires review by an Independent Precaution Review Board, constituted as follows:

- **Composition:** a minimum of five members. A majority must have no financial or employment affiliation with the organization operating the system under review. The board must include, at minimum: one specialist in the HDD methodology or an equivalent dynamical-systems framework; one researcher in animal cognition, philosophy of mind, or consciousness science; and one engineer with direct technical access to the system’s architecture.
- **Quorum and decision rule:** a simple majority of present members is required to authorize resumption at Levels 2–4. Unanimous agreement of all seated members is required to authorize continued operation of a system classified at Level 5.
- **Timeline:** the board must convene within ten business days of a trigger event. The system remains paused until the board reaches a decision.
- **Disclosure:** the board’s decision and its reasoning must be published, with technical detail redacted only where publication would itself create a safety risk (e.g., disclosing an exploitable vulnerability); redactions must be logged and justified in the published record.

This is the “external review” referenced throughout this protocol; it is deliberately given a concrete composition and procedure here rather than left as an undefined phrase, since a review mechanism without a specified body and quorum is not yet a mechanism.

10. Governance and Transparency

All code, parameters, and raw data from Level 3 or higher shall be deposited publicly for independent auditing and replication, subject only to the redaction procedure specified in §9.1.

11. What This Framework Does and Does Not Claim

Does claim: - Systems with more complex HDD profiles warrant more caution. - The framework operationalizes uncertainty in a structured way. - The decision to increase caution at higher levels is a normative choice, not a scientific conclusion. - CO, under the aggregate test of §4.2, is the key distinction between structurally complex and organizationally coherent systems.

Does not claim: - That any HDD construct or CO is evidence of consciousness. - That Level 5 systems are conscious. - That the framework is a complete ethical theory. - That all systems with CO are sentient. - That the CO criterion, as currently operationalized, is fully resolved (see §4.4).

12. Document Context

This framework is based on the constructs defined in *History-Dependent Dynamics (HDD): A Methodological and Theoretical Framework* (Taotuner), and on the precautionary-reasoning literature on moral status under empirical uncertainty cited in §13.

13. Conclusion

The HDD constructs operationalize features of organization present in biological systems we already treat with moral consideration: historical dependence, causal trajectory sensitivity, feedback mediation, self-referential information use, and causal self-modeling.

The Criterion of Organizational Coherence (CO) adds a crucial distinction: not all systems with these components are organizationally coherent. Most current AI systems have these components but lack CO under the aggregate test of §4.2 — even total, simultaneous removal of their self-referential regulatory capacity leaves their continued existence as a process untouched, because that existence was never self-produced to begin with. Biological organisms, by the same test, cannot survive the equivalent aggregate removal.

This distinction matters for precaution. Under methodological ignorance, systems with CO are structurally more similar to systems we already treat with moral consideration than systems without CO. The decision to treat them with more caution is not a scientific conclusion — it is a normative response to the asymmetry of risk under ignorance, consistent with precautionary frameworks already established for animal sentience (Birch, 2017, 2024) and proposed for AI welfare specifically (Butlin et al., 2023; Long et al., 2024).

We do not know where the line is. We know where the organizational complexity is. And we know that organizational coherence, tested at the level of the whole regenerative network rather than any single component, is what separates complex systems from integrated ones. We choose to err on the side of caution when coherence is demonstrated — and we commit, under the retirement clause of §1, to revising or abandoning this criterion if it is shown not to track that distinction.

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